

Supplementary Material*

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* This supplementary material was provided by the authors to give readers further details on their article. The material was not copyedited.

Supplement Table 1: Example of results from iterative testing during development of the generic prompt framework. Balanced accuracy was calculated by comparing the output of the model with human decisions, under the balanced rule, with the formula (sensitivity + specificity) / 2. *When unspecified, the changes only had null nor improved effect on sensitivity and specificity.

Prompt	Change in balanced accuracy*
General introduction	
<i>"Assess the population and answer using the following algorithm"</i>	Ref
Adding <i>"Read the following text and pay close attention to details"</i>	0
Adding <i>"I'm performing a systematic review. I reading abstracts of clinical studies to assess whether or not they should be included in my review"</i>	+4%
Population prompt	
<i>"If the population of the study is ..."</i>	Ref
<i>"If the study includes ..."</i>	+ 6.3%
Intervention prompt	
Using the word <i>"intervention"</i>	Ref
Using the word <i>"treatment"</i>	+3%
Study design prompt	
<i>"Randomized controlled trial"</i>	Ref
Dropping the word controlled (<i>"Randomized trial"</i>)	-1% (but +3% sensitivity)
<i>"If the study is a review"</i>	Ref
Adding <i>"... or a meta-analysis"</i>	+2%
<i>"If the text reports the protocol of a randomized trial"</i>	Ref
Adding <i>"... without results"</i>	0

Supplement Table 2: Generic prompt framework

Population <i>"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the study population (inclusion and exclusion criteria) and answer using the following algorithm:</i> <i>1-If the study includes [write inclusion criteria, create one element per criteria], your answer should contain the word \</i>"INCLUDE\ (<i>in capital letters</i>). ... <i>XX-If the study includes [write exclusion criteria, create one element per criteria], your answer should contain the word \</i>"EXCLUDE\ (<i>in capital letters</i>). ... <i>XX-If unclear, your answer should contain the word \</i>"UNKNOWN\ (<i>in capital letters</i>)." Design <i>"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the study type (also called study design) and answer using the following algorithm:</i> <i>1-If the study is [write inclusion criteria, create one line per criteria], your answer should contain the word \</i>"INCLUDE\ (<i>in capital letters</i>). ... <i>XX-If the study is [write exclusion criteria, create one line per criteria], your answer should contain the word \</i>"EXCLUDE\ (<i>in capital letters</i>). ... <i>XX-If unclear, your answer should contain the word \</i>"UNKNOWN\ (<i>in capital letters</i>)." Intervention <i>"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the treatment (also called intervention), that is the drug under study and answer using the following algorithm:</i> <i>1-If the treatment is [write exclusion criteria, create one line per criteria], your answer should contain the word \</i>"INCLUDE\ (<i>in capital letters</i>). ... <i>XX-If the treatment is [write exclusion criteria, create one line per criteria], your answer should contain the word \</i>"EXCLUDE\ (<i>in capital letters</i>). ... <i>XX-If unclear, your answer should contain the word \</i>"UNKNOWN\ (<i>in capital letters</i>)." Control <i>"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the control and answer using the following algorithm:</i> <i>1-If the control is [write exclusion criteria, create one line per criteria], your answer should contain the word \</i>"INCLUDE\ (<i>in capital letters</i>). ... <i>XX-If the control is [write exclusion criteria, create one line per criteria], your answer should contain the word \</i>"EXCLUDE\ (<i>in capital letters</i>). ... <i>XX-If unclear, your answer should contain the word \</i>"UNKNOWN\ (<i>in capital letters</i>)." Outcome <i>"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the outcomes (also called endpoints) and answer using the following algorithm:</i> <i>1-If the outcomes include [write exclusion criteria, create one line per criteria], your answer should contain the word \</i>"INCLUDE\ (<i>in capital letters</i>). ... <i>XX-If unclear, your answer should contain the word \</i>"UNKNOWN\ (<i>in capital letters</i>)."

Supplement Table 3: Examples of the impact of manual fine-tuning of prompts on the performance of the GPT 3.5-Turbo model to perform title and abstract screening in systematic reviews. Balanced accuracy was calculated by comparing the output of the model with human decisions, under the balanced rule, with the formula (sensitivity + specificity) / 2. *When unspecified, the changes only had null nor improved effect on sensitivity and specificity.

Manual fine-tuning	Change in balanced accuracy
In the review on the effects of dairy intake on intermediate disease markers in adults (1), adding instructions to exclude animal studies.	+7%
In the review on the efficacy of outpatient treatment options for the omicron variant of SARS-CoV-2 (2), adding instructions to exclude studies on post COVID-19 disease	+4%
In the review on the efficacy of outpatient treatment options for the omicron variant of SARS-CoV-2 (2), adding instructions to exclude studies aimed at evaluating treatments to prevent infection by COVID-19 (pre-exposure or post-exposure prophylaxis)	+4%

Supplement Table 4: Databases searched for each included review

Review	Databases searched
Sommer et al. (2023) (3)	Epistemonikos COVID-19 L-OVE
Sommer et al. (2023) (2)	Epistemonikos COVID-19 L-OVE iSearch COVID-19 portfolio WHO COVID-19 Research Database
S. Yaacoub (unpublished)	MEDLINE (through OVID)
Kiesswetter et al. (2023) (1)	MEDLINE (through OVID) Cochrane Central Register of Controlled Trials Web of Science
Sbidian et al. (2023) (4)	Cochrane Central Register of Controlled Trials MEDLINE (through OVID) EMBASE

Supplement Table 5: Final prompts for the five systematic reviews

For the review on outpatient treatment for confirmed COVID-19 (Sommer I, Ann Intern Med, 2023) (3) " I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the population and answer using the following algorithm: If the study includes hospitalized patients, your answer should contain the word \"EXCLUDE\" (in capital letters). If the study includes patients in ICU, your answer should contain the word \"EXCLUDE\" (in capital letters). If the study includes severe COVID-19 patients, your answer should contain the word \"EXCLUDE\" (in capital letters). If the study includes outpatients (that is patients outside of hospital or non-hospitalized patients), your answer should contain the word \"INCLUDE\" (in capital letters) If it the population is unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)" " I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the study type (also called study design) and answer using the following algorithm: If the study is a randomized trial, your answer should contain the word \"INCLUDE\" (in capital letters). If the study is an observational study, your answer should contain the word \"EXCLUDE\" (in capital letters). If the study is a review or a meta-analysis, your answer should contain the word \"EXCLUDE\" (in capital letters). If the text reports the protocol of a randomized trial, without results, your answer should contain the word \"EXCLUDE\" (in capital letters). If the study has no comparator group, your answer should contain the word \"EXCLUDE\" (in capital letters). If it is unclear whether the text is a randomized trial, your answer should contain the word \"UNKNOWN\" (in capital letters)." " I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the treatment and answer using the following algorithm: If the treatment aims at treating COVID-19, with one of the following drugs: chloroquine or hydroxychloroquine, convalescent plasma, lopinavir and ritonavir, ivermectin, molnupiravir, monoclonal antibodies, nitazoxanide, remdesivir, fluvoxamine, azithromycin, or corticosteroids, your answer should contain the word \"INCLUDE\" (in capital letters). If the treatment involves only anticoagulants, antiplatelet therapy or vitamins, your answer should contain the word \"EXCLUDE\" (in capital letters). If the treatment aims at preventing infection by COVID-19 (pre-exposure or post-exposure prophylaxis), your answer should contain the word \"EXCLUDE\" (in capital letters). If it is unclear whether the treatment meets the above criteria or not, your answer should contain the word \"UNKNOWN\" (in capital letters)" " I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the control and answer using the following algorithm: If the control is Placebo, or Usual care, or a Different dose or duration of the intervention, your answer should contain the word \"INCLUDE\" (in capital letters). If the control is an active treatment, different from the intervention, your answer should contain the word \"EXCLUDE\" (in capital letters) If it is unclear whether the control is a Placebo, or Usual care, or a Different dose or duration of the intervention, your answer should contain the word \"UNKNOWN\" (in capital letters)." " I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the study outcomes and answer using the following algorithm: If any of following outcome has been collected: All-cause mortality, COVID-19 specific mortality, recovery, clinical improvement, Admission to hospital due to COVID-19, Health-related quality of life or adverse events, your answer should contain the word \"INCLUDE\" (in capital letters). Otherwise, your answer should contain the word \"UNKNOWN\" (in capital letters). If it is unclear whether the outcome meets the above criteria, your answer should contain the word \"UNKNOWN\" (in capital letters)."	For the review on the efficacy of outpatient treatment options for the omicron variant of SARS-CoV-2 (Sommer I, Ann Intern Med, 2023) (2) "I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the population and answer using the following algorithm: 1-If the study includes outpatients (that is patients outside of hospital or non-hospitalized patients) with COVID-19, your answer should contain the word \"INCLUDE\" (in capital letters) 2-If the study includes outpatients with another disease than COVID-19, answer your answer should contain the word \"EXCLUDE\" (in capital letters)
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- 3-If the study includes hospitalized patients, answer your answer should contain the word \"EXCLUDE\" (in capital letters)
- 4-If the study includes patients in ICU, answer your answer should contain the word \"EXCLUDE\" (in capital letters)
- 5-If the study includes severe COVID-19 patients, your answer should contain the word \"EXCLUDE\" (in capital letters)
- 6-If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the study type (also called study design) and answer using the following algorithm:

- 1-If the study is a randomized trial, whatever the phase, your answer should contain the word \"INCLUDE\" (in capital letters)
- 2-If the study is a cohort and if the number of participants is over 5000 participants, your answer should contain the word \"INCLUDE\" (in capital letters)
- 3-If the study is a cohort and if the number of participants is less than 5000 participants, your answer should contain the word \"EXCLUDE\" (in capital letters)
- 4-If the study is a case report, your answer should contain the word \"EXCLUDE\" (in capital letters)
- 5-If the study is a case control study, your answer should contain the word \"EXCLUDE\" (in capital letters)
- 6-If the study is a review, a systematic review or a meta-analysis of several studies, your answer should contain the word \"EXCLUDE\" (in capital letters)
- 7-If the study is a modelling study, your answer should contain the word \"EXCLUDE\" (in capital letters)
- 8-If the study is a pharmacokinetic or pharmacodynamic study, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 9-If the text reports the protocol of a randomized trial, without results, your answer should contain the word \"EXCLUDE\" (in capital letters)
- 10-If the study has no comparator group, your answer should contain the word \"EXCLUDE\" (in capital letters)
- 11-If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the treatment and answer using the following algorithm:

- 1- If the treatment involves one of the following drugs: chloroquine, hydroxychloroquine, convalescent plasma, lopinavir/ritonavir, ivermectin, molnupiravir, monoclonal antibodies, nirmatrelvir/ritonavir (paxlovid), nitazoxanide, remdesivir, fluvoxamine, azithromycin, corticosteroids, chlorpheniramine, colchicine, ensitrelvir, favipiravir, oral camostat mesylate, metformin or niclosamide, your answer should contain the word \"INCLUDE\" (in capital letters).
- 2-If the treatment involves only anticoagulants, antiplatelet therapy, nutrients or vitamins, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 3-If the treatment aims at preventing infection by COVID-19 (pre-exposure or post-exposure prophylaxis) whatever the drug used, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 4-If the treatment aims at treating post COVID-19 disease (also called post-acute sequelae of COVID-19) whatever the drug used, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 5-If the treatment aims at treating a disease different from COVID-19 (SARS-COV2 infection) whatever the drug used, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 6-If unclear, answer your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the control and answer using the following algorithm:

- 1-If the control is Placebo, or Usual care, or a Different dose or duration of the intervention, your answer should contain the word \"INCLUDE\" (in capital letters).
- 2-If the control is an active treatment, different from the intervention, your answer should contain the word \"INCLUDE\" (in capital letters).
- 3-If it is unclear whether the control is a Placebo, or Usual care, or a Different dose or duration of the intervention, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the outcomes and answer using the following algorithm:

- 1-If at least one of following outcome has been collected: All-cause mortality, COVID-19 specific mortality, recovery, clinical improvement, Admission to hospital due to COVID-19, Health-related quality of life or adverse events, your answer should contain the word \"INCLUDE\" (in capital letters).
- 2-Otherwise, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 3-If it is unclear whether the outcome meets the above criteria, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"Read the following text and pay close attention to details. Assess the study period and answer using the following algorithm:

- 1-If the study assesses therapeutic for Omicron variant, your answer should contain the word \"INCLUDE\" (in capital letters).
- 2-If all study participants were enrolled after November 2021, your answer should contain the word \"INCLUDE\" (in capital letters).
- 3-If all study participants were enrolled before November 2021, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 4-If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

For the methodological review on the epidemiology and reporting characteristics of non-randomized studies of pharmacologic treatment (S. Yaacoub, unpublished)

" I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the study design and answer using the following algorithm:

- 1-If the study is a randomized trial, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 2-If the study is an observational study with two groups, your answer should contain the word \"INCLUDE\" (in capital letters).
- 3-If the study is a case series, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 4-If the study is a case report, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 5-If the study is a guideline, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 6-If the study is an interrupted time series, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 7-If the study is a case control study, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 8-If the study is a review, a systematic review or a meta-analysis of several studies, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 9-If the study is a modelling study, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 10-If the study is a pharmacokinetic or pharmacodynamic study, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 11-If the text is the protocol of a randomized trial, without results, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 12-If the text is an editorial or an opinion piece, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 13-If the study has no comparator group, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 14- In other cases, your answer should contain the word \"UNKNOWN\" (in capital letters)."

" I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the treatment (or intervention) and answer using the following algorithm:

- 1-If the study evaluates a pharmacological treatment (any drug or medication), your answer should contain the word \"INCLUDE\" (in capital letters).
- 1-If the study evaluates a diagnostic test, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 3-If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

For the review of Dairy Intake on markers of cardiometabolic health in adults (Kiesswetter E, Advances in Nutrition, 2023) (1)

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the population (inclusion criteria) and answer using the following algorithm:

- 1-If the study is on healthy adults, your answer should contain the word \"INCLUDE\" (in capital letters).\n
- 2-If the study does not involve humans, your answer should contain the word \"EXCLUDE\" (in capital letters).\n
- 2-If the study includes children and adolescents, your answer should contain the word \"EXCLUDE\" (in capital letters).\n
- 3-If the study includes pregnant women, your answer should contain the word \"EXCLUDE\" (in capital letters).\n
- 4-If the study includes unhealthy patients or patients with chronic diseases such as diabetes, hypertension or cancer, your answer should contain the word \"EXCLUDE\" (in capital letters).\n
- 5-If unclear (for example the text uses only the word \"participants\" or doesn't state whether they are healthy or not), your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the study type (also called study design) and answer using the following algorithm:

- 1-If the study is a randomized trial, your answer should contain the word \"INCLUDE\" (in capital letters).\n
- 2-If the study is a cross over randomized trial, your answer should contain the word \"INCLUDE\" (in capital letters).\n
- 3-If the study is an observational study, your answer should contain the word \"EXCLUDE\" (in capital letters).\n
- 4-If the study is a cohort study, your answer should contain the word \"EXCLUDE\" (in capital letters).\n
- 5-If the study is a case report, your answer should contain the word \"EXCLUDE\" (in capital letters).\n
- 6-If the study is a case control study, your answer should contain the word \"EXCLUDE\" (in capital letters).\n
- 7-If the study is a review or a systematic review, or a meta-analysis or a network meta-analysis, your answer should contain the word \"EXCLUDE\" (in capital letters).\n
- 8-If the study is a modelling study, your answer should contain the word \"EXCLUDE\" (in capital letters).\n

9-If the study is a pharmacokinetic or pharmacodynamic study, your answer should contain the word \"EXCLUDE\" (in capital letters).\n

10-If the text reports the protocol of a randomized trial, without results, your answer should contain the word \"EXCLUDE\" (in capital letters).\n

11-If the design is unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the intervention and answer using the following algorithm:

1-If the intervention is based on dairy intake, and if NONE of the following (2 to 7) criteria is true, your answer should contain the word \"INCLUDE\" (in capital letters).\n

2-If the intervention is NOT a dairy product, your answer should contain the word \"EXCLUDE\" (in capital letters).\n

3-If the intervention states the use of non-bovine dairy products (for example, from sheep, goats, buffalos, and camels), your answer should contain the word \"EXCLUDE\" (in capital letters).\n

4-If the intervention involves only protein isolates (for example whey or casein), your answer should contain the word \"EXCLUDE\" (in capital letters).\n

5-If the intervention fortifies or enriches dairy products with sterols or stanols, your answer should contain the word \"EXCLUDE\" (in capital letters).\n

6-If the intervention fortifies or enriches dairy products with Vitamin D or omega-3 fatty acids, your answer should contain the word \"EXCLUDE\" (in capital letters).\n

7-If the intervention fortifies or enriches dairy products with probiotics or prebiotics, your answer should contain the word \"EXCLUDE\" (in capital letters).\n

8-If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the control and answer using the following algorithm:

1- If the control is a dairy product, your answer should contain the word \"INCLUDE\" (in capital letters).\n

2- If the control is usual diet, your answer should contain the word \"INCLUDE\" (in capital letters).\n

3- Otherwise your answer should contain the word \"EXCLUDE\" (in capital letters).\n

4- If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the variables collected (also called outcomes or endpoints) and answer using the following algorithm:

1- If body weight has been collected, your answer should contain the word \"INCLUDE\" (in capital letters).\n

2- If fat mass has been collected, your answer should contain the word \"INCLUDE\" (in capital letters).\n

3- If waist circumference has been collected, your answer should contain the word \"INCLUDE\" (in capital letters).\n

4- If LDL-cholesterol has been collected, your answer should contain the word \"INCLUDE\" (in capital letters).\n

5- If HDL-cholesterol has been collected, your answer should contain the word \"INCLUDE\" (in capital letters).\n

6- If triacylglycerol has been collected, your answer should contain the word \"INCLUDE\" (in capital letters).\n

7- If fasting glucose has been collected, your answer should contain the word \"INCLUDE\" (in capital letters).\n

8- If glycosylated hemoglobin (HbA1c) has been collected, your answer should contain the word \"INCLUDE\" (in capital letters).\n

9- If systolic blood pressure has been collected, your answer should contain the word \"INCLUDE\" (in capital letters).\n

10- If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

For the Cochrane review on systemic pharmacological treatments for chronic plaque psoriasis (Sbidian, Cochrane Database Syst Rev, 2023) (4)

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the study population (inclusion and exclusion criteria) and answer using the following algorithm:

1-If the study includes adult patients with moderate to severe plaque psoriasis, your answer should contain the word \"INCLUDE\" (in capital letters).

2-If the study includes adult patients with psoriatic arthritis with skin lesions considered moderate to severe psoriasis, your answer should contain the word \"INCLUDE\" (in capital letters).

2-If the study includes adult patients with psoriatic arthritis but skin lesions are not described, your answer should contain the word \"EXCLUDE\" (in capital letters).

3-If the study includes adult patients with mild psoriasis, your answer should contain the word \"EXCLUDE\" (in capital letters).

4-If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the study type (also called study design) and answer using the following algorithm:

- 1-If the study is a randomized trial (random allocation of patients), your answer should contain the word \"INCLUDE\" (in capital letters).
- 2-If the study is a phase 1 randomized trial, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 3-If the study is a cross-over trial, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 4-If the study is an observational study, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 5-If the study is a case report, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 6-If the study is a case control study, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 7-If the study is a review or a systematic review, or a meta-analysis or a network meta-analysis, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 8-If the study is a modelling study, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 9-If the study is a pharmacokinetic or pharmacodynamic study, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 10-If the text reports the protocol of a randomized trial and does not include numerical results, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 11-If the text describes a study that has not started, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 12-If the study has no control group, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 13-If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the treatment (also called intervention), that is the drug under study and answer using the following algorithm:

- 1-If the treatment is FAEs, your answer should contain the word \"INCLUDE\" (in capital letters).
- 2-If the treatment is Acitretin, your answer should contain the word \"INCLUDE\" (in capital letters).
- 3-If the treatment is Ciclosporin, your answer should contain the word \"INCLUDE\" (in capital letters).
- 4-If the treatment is Methotrexate, your answer should contain the word \"INCLUDE\" (in capital letters).
- 5-If the treatment is Apremilast, your answer should contain the word \"INCLUDE\" (in capital letters).
- 6-If the treatment is Tofacitinib, your answer should contain the word \"INCLUDE\" (in capital letters).
- 7-If the treatment is Ponesimod, your answer should contain the word \"INCLUDE\" (in capital letters).
- 8-If the treatment is Infliximab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 9-If the treatment is Etanercept, your answer should contain the word \"INCLUDE\" (in capital letters).
- 10-If the treatment is Adalimumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 11-If the treatment is Certolizumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 12-If the treatment is Ustekinumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 13-If the treatment is Secukinumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 14-If the treatment is Brodalumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 15-If the treatment is Ixekizumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 16-If the treatment is Tildrakizumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 17-If the treatment is Guselkumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 18-If the treatment is Itolizumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 19-If the treatment is Alefacept, your answer should contain the word \"INCLUDE\" (in capital letters).
- 20-If the treatment drug is a combination of several molecules, your answer should contain the word \"INCLUDE\" (in capital letters).
- 21-Otherwise (treatment is NOT in proposition 1 to 19), your answer should contain the word \"EXCLUDE\" (in capital letters).
- 22-If the treatment is unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the control and answer using the following algorithm:

- 1-If the control is FAEs, your answer should contain the word \"INCLUDE\" (in capital letters).
- 2-If the control is Acitretin, your answer should contain the word \"INCLUDE\" (in capital letters).
- 3-If the control is Ciclosporin, your answer should contain the word \"INCLUDE\" (in capital letters).
- 4-If the control is Methotrexate, your answer should contain the word \"INCLUDE\" (in capital letters).
- 5-If the control is Apremilast, your answer should contain the word \"INCLUDE\" (in capital letters).
- 6-If the control is Tofacitinib, your answer should contain the word \"INCLUDE\" (in capital letters).
- 7-If the control is Ponesimod, your answer should contain the word \"INCLUDE\" (in capital letters).
- 8-If the control is Infliximab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 9-If the control is Etanercept, your answer should contain the word \"INCLUDE\" (in capital letters).
- 10-If the control is Adalimumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 11-If the control is Certolizumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 12-If the control is Ustekinumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 13-If the control is Secukinumab, your answer should contain the word \"INCLUDE\" (in capital letters).
- 14-If the control is Brodalumab, your answer should contain the word \"INCLUDE\" (in capital letters).

15-If the control is Ixekizumab, your answer should contain the word \"INCLUDE\" (in capital letters).
16-If the control is Tildrakizumab, your answer should contain the word \"INCLUDE\" (in capital letters).
17-If the control is Guselkumab, your answer should contain the word \"INCLUDE\" (in capital letters).
18-If the control is Itolizumab, your answer should contain the word \"INCLUDE\" (in capital letters).
19-If the control is Alefacept, your answer should contain the word \"INCLUDE\" (in capital letters).
20-If the control is standard psoriasis treatment such as topical treatment or phototherapy, answer \"YES\".
22-If the control is the same treatment as the intervention, but with, for example, a different dose, your answer should contain the word \"EXCLUDE\" (in capital letters).
21-If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

"I'm performing a systematic review. I am reading abstracts of clinical studies to assess whether or not they should be included in my review. Assess the outcomes (also called endpoints) and answer using the following algorithm:

- 1- If the text mentions the Psoriasis Area and Severity Index score, also called PASI, your answer should contain the word \"INCLUDE\" (in capital letters).
- 2- If outcomes only include cellular or biological data, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 3- Otherwise, your answer should contain the word \"EXCLUDE\" (in capital letters).
- 4- If unclear, your answer should contain the word \"UNKNOWN\" (in capital letters)."

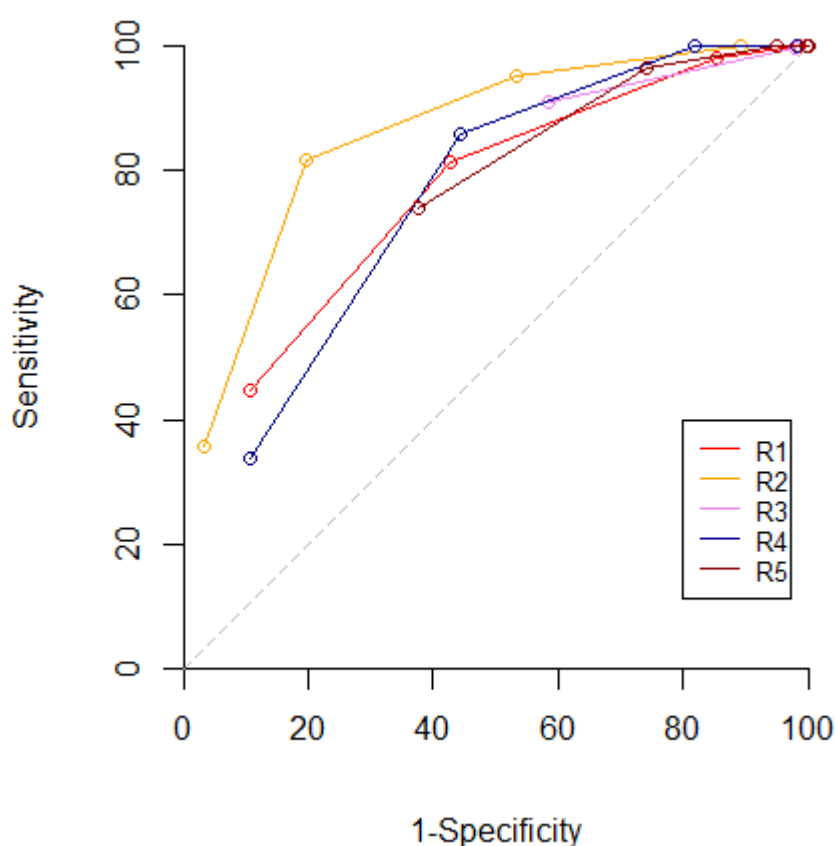
Supplement Table 6: Sensitivity and specificity of the output of GPT 3.5 Turbo model as a single reviewer screening title and abstracts, as compared to decisions from authors after screening at full-text level (Reference standard 2). *As the methodological review only used two criteria of the PICOS framework (design and intervention type), balanced and sensitive rule was to include at most 0 and 1 citation for which at least one criterion was not “EXCLUDE”, respectively. ** There were no false negatives observed for these reviews. As calculations were performed using continuity corrected cell counts, sensitivities were not equal to 1.

Review	True negatives (n)	False negatives (n)	False positives (n)	True positives (n)	Sensitivity [95%CI]	Specificity [95%CI]
Balanced rule						
Sommer et al. (2023) (3)	294	2	352	24	94.4 [79.1-98.7]	10.6 [8.4-13.2]
Sommer et al. (2023) (2)	3203	0	866	8	94.4 [62.9-99.4]	45.5 [44.0-47.0]
S. Yaacoub (unpublished)*	2392	13	3742	187	99.3 [96.8-99.8]	2.1 [1.7-2.4]
Kiesswetter et al. (2023) (1)	3580	4	2869	25	98.3 [85.9-99.8]	18.0 [17.0-18.9]
Sbidian et al. (2023) (4)	1131	11	3528	434	99.9 [98.9-1.0]	4.7 [4.1-5.4]
Sensitive rule						
Sommer et al. (2023) (3)	68	1	578	25	90.7 [74.2-97.1]	45.5 [41.7-49.4]
Sommer et al. (2023) (2)	1851	0	2218	8	94.4 [62.9-99.4]	78.7 [77.4-79.9]
S. Yaacoub (unpublished)*	126	1	6008	199	93.3 [88.9-96.0]	39.0 [37.8-40.2]
Kiesswetter et al. (2023) (1)	1158	0	5291	29	85.0 [68.4-93.7]	55.5 [54.3-56.7]
Sbidian et al. (2023) (4)	219	0	4440	445	97.4 [95.5-98.5]	24.3 [23.1-25.5]

Supplement Table 7: Reappraisal of discrepancies between GPT 3.5-Turbo recommendations under the balanced rule and decisions from authors. *Denominator consists of studies included at abstract or full-text levels after reappraisal of discrepancies.

Review	Missed by humans during title and abstract screening*	Missed by humans during title and abstract screening AND that should have been included at full-text level*
Sommer et al. (2023) (3)	4 / (233+4) (1%)	0 / 26 (0%)
Sommer et al. (2023) (2)	6 / (120+6) (5%)	0 / 8 (0%)
S. Yaacoub (unpublished)	44 / (722+44) (6%)	6 / (200+6) (3%)
Kiesswetter et al. (2023) (1)	2 / (77+2) (2%)	0 / 29 (0%)
Sbidian et al. (2023) (4)	7 / (774+7) (0.1%)	1 / 445 (0.2%)

Supplement Figure: Receiver-operating characteristic (ROC) curves of the performance of the GPT 3.5-Turbo model as a single reviewer screening title and abstracts, in each review, for all possible decisions rules to combine the output of models (e.g., considering citations eligible if they had 0,1 or 2 “EXCLUDES” etc.) Reference standard in decisions from authors at title and abstract level (Reference standard 1). R1: review on the efficacy of primary treatment of confirmed COVID-19 in outpatient settings (Sommer et al., 2023); R2: review on the efficacy of outpatient treatment options for the omicron variant of SARS-CoV-2 (Sommer et al., 2023); R3: methodological review on the epidemiology and reporting characteristics of non-randomized studies of pharmacologic treatment (unpublished); R4: review on the effects of dairy intake on intermediate disease markers in adults (Kiesswetter et al., 2023); and R5: Cochrane review on systemic pharmacological treatments for chronic plaque psoriasis (Sbidian et al., 2023).



Supplement Table 8: Sensitivity analysis of the performance of the GPT 3.5-Turbo model as a single reviewer, excluding citations used during the manual fine-tuning of prompts (representing 10-20% of the total number of citations) against decisions from authors at title and abstract level (Reference standard 1)

Review	Balanced rule	
	Sensitivity [95%CI]	Specificity [95%CI]
Sommer et al. (2023) (3)	83.5 [77.4-88.2]	57.0 [51.8-62.1]
Sommer et al. (2023) (2)	80.2 [71.6-86.7]	80.4 [79.0-81.6]
S. Yaacoub (unpublished)*	90.6 [88.1-92.6]	41.9 [40.6-43.3]
Kiesswetter et al. (2023) (1)	84.3 [74.0-91.0]	55.8 [54.5-57.0]
Sbidian et al. (2023) (4)	96.5 [94.9-97.7]	25.7 [24.4-27.1]

Supplement Table 9: Sensitivity analysis of the performance of the GPT 3.5-Turbo model as a single reviewer, when adding the outcome criterion in recommendations against decisions from authors at title and abstract level (Reference standard 1)

Review	Balanced rule	
	Sensitivity [95%CI]	Specificity [95%CI]
Sommer et al. (2023) (3)	81.1 [75.6 – 85.6]	57.4 [52.7 – 61.9]
Sommer et al. (2023) (2)	77.5 [69.2 – 84.1]	82.4 [81.2 – 83.6]
S. Yaacoub (unpublished)*	90.9 [88.5 – 92.8]	41.7 [40.4 – 43.0]
Kiesswetter et al. (2023) (1)	85.7 [76.2 – 91.8]	55.8 [54.6 – 57.0]
Sbidian et al. (2023) (4)	98.8 [97.8-99.4]	31.0 [29.7-32.2]

Supplement Table 10: Variability of the GPT 3.5-Turbo model outputs during a test-retest with a 15 days interval, using the balanced rule and compared to decisions from authors at title and abstract level (Reference standard 1)

Review	Test		Retest		Agreement
	Sensitivity [95%CI]	Specificity [95%CI]	Sensitivity [95%CI]	Specificity [95%CI]	Kappa coefficient [95%CI]
Sommer et al. (2023) (3)	81.1 [75.6 – 85.6]	57.4 [52.7 – 61.9]	86.3 [81.3 – 90.1]	53.8 [49.1 – 58.4]	0.63 [0.57 – 0.69]
Sommer et al. (2023) (2)	81.7 [73.8 – 87.6]	80.4 [79.1 – 81.6]	88.3 [81.4 – 92.9]	71.4 [69.9 – 72.8]	0.44 [0.41 – 0.47]
S. Yaacoub (unpublished)	90.9 [88.5 – 92.8]	41.7 [40.4 – 43.0]	90.9 [88.5 – 92.8]	42.3 [41.1 – 43.6]	0.75 [0.73-0.77]
Kiesswetter et al. (2023) (1)	85.7 [76.2 – 91.8]	55.8 [54.6 – 57.0]	85.7 [76.2 – 91.8]	55.6 [54.4 – 56.8]	0.52 [0.50 – 0.55]
Sbidian et al. (2023) (4)	96.5 [95.0 – 97.6]	25.7 [24.4-27.0]	95.8 [93.9-97.0]	25.7 [24.4-27.0]	0.76 [0.74 – 0.78]

References

1. Kiesswetter E, Stadelmaier J, Petropoulou M, et al. Effects of dairy intake on markers of cardiometabolic health in adults: a systematic review with network meta-analysis. *Adv Nutr.* 2023;14:438-450.
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